

Cody Williams · UTA · Age 21

97.9%

breakout probability next season

Could we have predicted the breakout of Jalen Brunson? I trained an ML model on **15 years of NBA data** to find the next one. ↓

Swipe for the full list + methodology →

Top 10 Breakout Candidates

1	Cody Williams	UTA		97.9%
2	Stephon Castle	SAS		92.2%
3	Anthony Black	ORL		91.8%
4	Bilal Coulibaly	WAS		88.7%
5	Bub Carrington	WAS		87.7%
6	Kyshawn George	WAS		87.6%
7	Zaccharie Risacher	ATL		83.5%
8	Brice Sensabaugh	UTA		82.3%
9	Brandin Podziemski	GSW		82.0%
10	Ja'Kobe Walter	TOR		80.6%

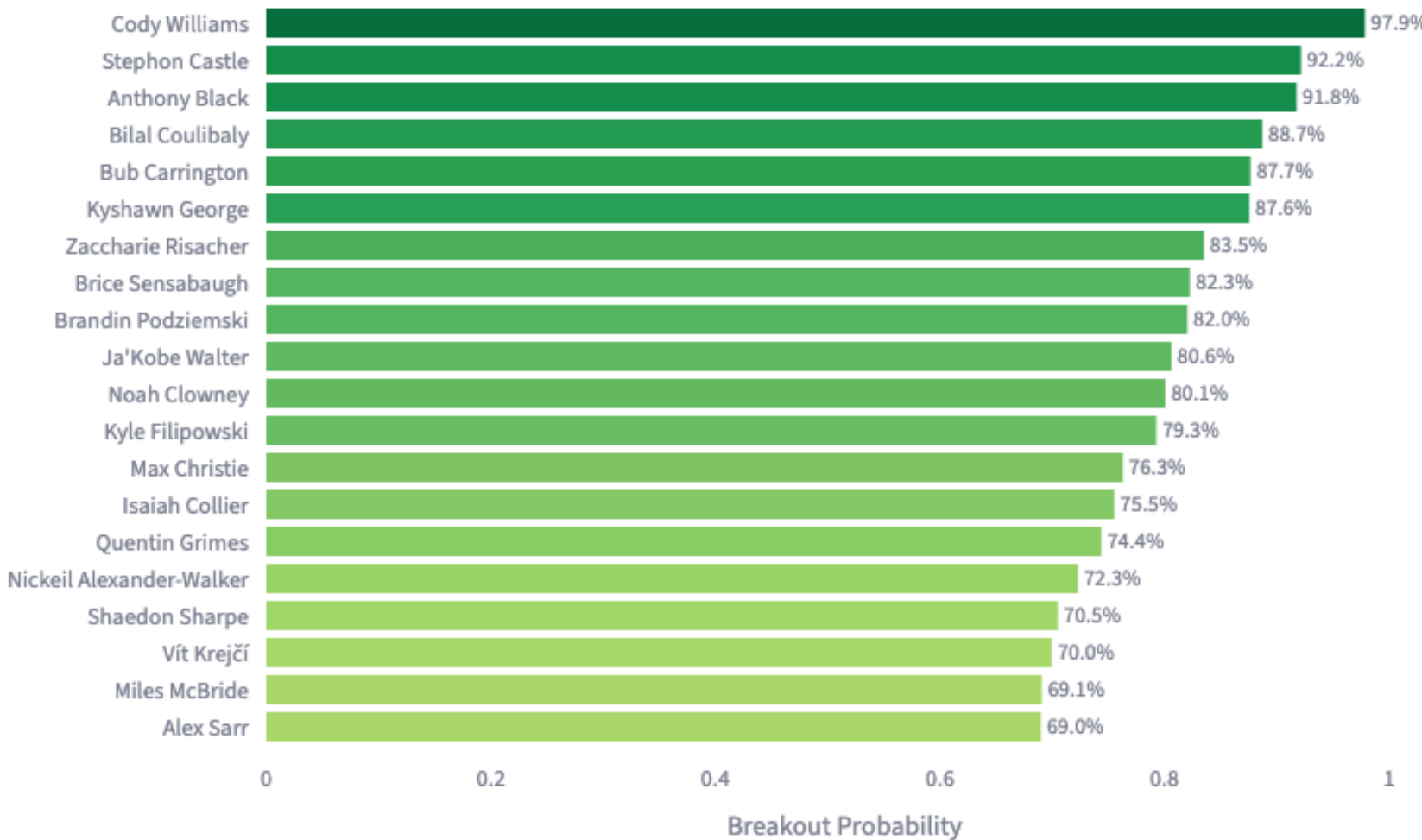
Breakout = ≥20% improvement in PTS, PIE, or TS% · Minimum 40 GP, 20 MPG both seasons · Trained on 2010–2025 data



2027 Breakout Candidates

Players predicted to have a breakout season (≥20% jump in PTS, PIE, or TS%) in 2026-27.

Top Breakout Candidates — 2026-27



Full Table

Player	Team	Age	PPG (25-26)	TS% (25-26)	USG% (25-26)	PIE (25-26)
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Player Deep Dive

Select a player

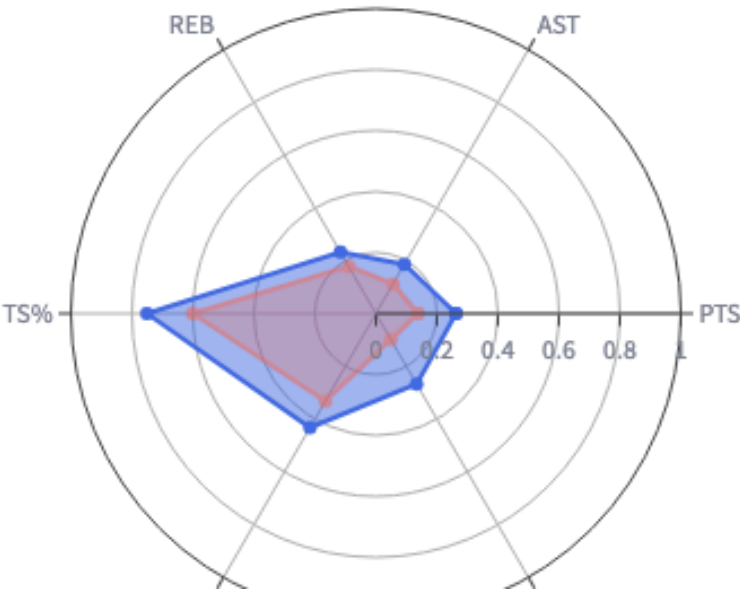
Cody Williams

Breakout Prob	Age	Team	Years in League
97.9%	21.0	UTA	1

Stat Radar: 2025-26 vs Prior Season

>  What do these stats mean?

Cody Williams — Normalized Stats



🔍 Player Deep Dive

Select a player

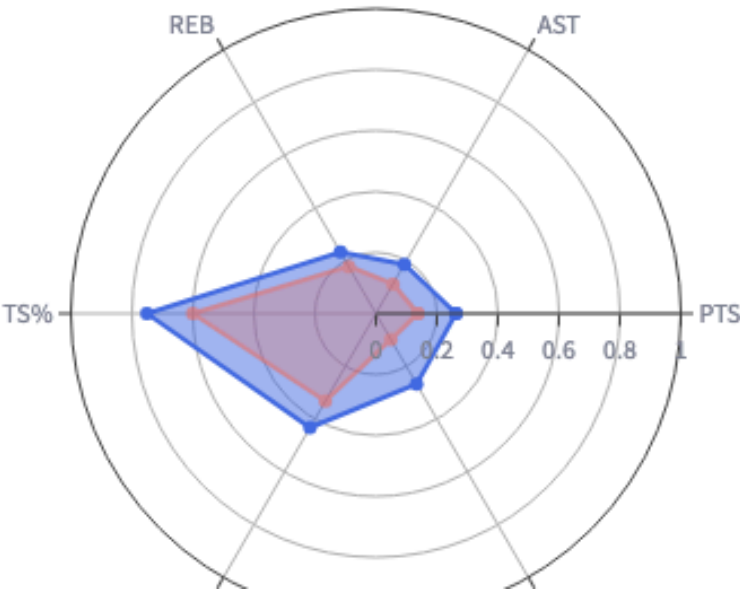
Cody Williams

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Stat Radar: 2025-26 vs Prior Season

> 📖 What do these stats mean?

Cody Williams — Normalized Stats





Model Performance

Evaluated on 2025-26 held-out validation season.

>  How to read these results

Model	Precision	Recall	F1	AUC-ROC	Avg Precision
XGBoost	0.500	0.162	0.245	0.677	0.477
Random Forest	0.550	0.297	0.386	0.717	0.496
★ LightGBM	0.562	0.243	0.340	0.655	0.439

Model Comparison — Validation Season 2025-26



How the model works

1

Data — 15 seasons from the NBA API

Per-game + advanced stats · SQLite · JSON caching with retry logic

~30K rows

2

Feature Engineering — 60+ signals per player

Prior-season stats · YoY deltas · Age · Years in league · Team change

No leakage

3

3 Models trained + compared

XGBoost · Random Forest · LightGBM · Threshold tuned for precision

56% precision

4

SHAP — every prediction explained

TreeExplainer · Top 8 drivers per player · Not a black box

4× baseline

5

Streamlit dashboard — deployed free

Plotly charts · Filter by age · Player deep dive · Model performance

Live now

OPEN SOURCE · FREE · LIVE

Built with Python & ML. Open source. Try it.

56%

Precision · 4× baseline

15

Seasons of training data

60+

Engineered features

20

Breakout candidates ranked

 nba-breakout-predictor.streamlit.app

 github.com/ChenEylon/NBA-breakout-predictor